

STRICT IMPLICATION

Historical preamble

Modal logic was discussed by several ancient authors, notably Aristotle,¹ and also by mediaeval logicians; their work, however, lies outside the scope of this book. The subject then appears to have been almost completely neglected until fairly recent times. In fact the first steps towards modern modal logic seem to have been taken by Hugh MacColl towards the end of the 19th century. MacColl introduces the operations of disjunction ($a + b$), negation (a') and implication ($a : b$).² He then asserts as a valid principle

$$(a : b) : a' + b$$

but denies the validity of

$$(a : b) = a' + b$$

on the ground that if a means 'He will persist in his extravagancy' and b means 'He will be ruined', then the negation of $a : b$ is 'He may persist in his extravagancy without necessarily being ruined', while the negation of $a' + b$ is 'He will persist in his extravagancy and he will not be ruined'. MacColl objects to the identification of these precisely because the first asserts only possibility while the second asserts something more. What this amounts to is that he regards $a : b$ as expressing necessary implication, and $a' + b$ as expressing material implication. In later papers, and in his book entitled *Symbolic Logic and its Applications*, this becomes even clearer: for he explicitly denies that his implicational

connective can be given a truth-functional interpretation, and he defines $(A : B)$ as $(A' + B)^{\epsilon}$ (or alternatively as $(AB')^{\eta}$), where ϵ and η represent necessity and impossibility respectively.³

But MacColl does not give any axioms⁴ and his system can hardly be called a modal logic of the distinctively modern kind with which this book is concerned. For that we have to wait until shortly after the publication in 1910 of *Principia Mathematica*,⁵ a work which did more than any other to establish the axiomatic method in logic. Beginning in 1912 C.I. Lewis published a series of articles and books⁶ in which he expressed dissatisfaction with the notion of material implication found in *Principia*. The grounds of his dissatisfaction were very much the same as those of MacColl, but he had the great advantage of being able to use an axiomatic method based on that of *Principia* itself, and he used it to construct a system (or rather a series of systems) in which material implication no longer played the dominant role. It is the work of Lewis which marks the beginning of modern modal logic properly so called.

The 'paradoxes of implication'

In the system of *Principia Mathematica* — indeed in any standard system of PC — there are found the theorems:

- (1) $p \supset (q \supset p)$
- (2) $\sim p \supset (p \supset q)$

The sense of (1) is often expressed by saying that if a proposition is true, any proposition whatsoever implies it: that of (2) by saying that if a proposition is false, it implies any proposition whatsoever. Together they are often called the 'paradoxes of (material) implication'. Moreover, since for any proposition p , either the antecedent of (1) or the antecedent of (2) must be true, it is easy to derive from (1) and (2) the further theorem:

- (3) $(p \supset q) \vee (q \supset p)$

i.e. in any pair of propositions, either the first implies the second or the second implies the first.

Lewis did not wish to reject these theorems. On the contrary, he argued (and surely correctly) that (1) and (2), when properly understood, are 'neither mysterious sayings, nor great discoveries, nor gross absurdities', but merely reflect the truth-functional sense in which Whitehead and Russell were using the word 'imply'. But he also

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maintained that there is another, stronger, sense of 'imply', a sense in which when we say that p implies q we mean that q follows from p ; and that in this sense of 'imply' it is not the case that every true proposition is implied by any proposition whatsoever, nor that every false proposition implies any proposition whatsoever. Moreover in this stronger sense of 'imply' there are pairs of propositions neither of which implies the other. Lewis was thus led to draw the distinction between an implication which holds materially and one which holds necessarily or *strictly*,⁷ and to make analogous distinctions for disjunction and equivalence. Before examining Lewis's modal logic we shall have something to say about the relation between propositions that he was attempting to capture.

The symbol Lewis used for strict implication was $\rightarrow$, and he interpreted $p \rightarrow q$ to mean that it is impossible that p should be true without q 's being true too. An alternative way of expressing the fact that it is impossible for p to be true without q also being true is to say that it is necessary that if p is true so is q , i.e. that $L(p \supset q)$ is true. In view of this equivalence we shall not have to take $\rightarrow$ as primitive but can define it as follows:

[Def $\rightarrow$] $(\alpha \rightarrow \beta) =_{\text{df}} L(\alpha \supset \beta)$

If instead of L we had taken M as primitive we could have defined $\alpha \rightarrow \beta$ as $\sim M(\alpha \wedge \sim \beta)$.

When two propositions strictly imply each other we say that each is *strictly equivalent* to the other. We use $=$ as the strict equivalence sign and introduce it by the definition:

[Def $=$] $(\alpha = \beta) =_{\text{df}} ((\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha))$ ⁸

Material and strict implication

It is not hard to see how replacing $\supset$ with $\rightarrow$ affects formulae like (1)–(3) on p. 194. Important differences between strict and material implication can be brought out, even in the system K, by comparing certain pairs of formulae. Sometimes a formula containing occurrences of $\supset$ is a theorem, but when $\supset$ is replaced by $\rightarrow$ the formula ceases to be a theorem. (Of course in any normal system this will never be the case when the only occurrence of $\supset$ so replaced is the main operator, for then either both formulae are theorems or neither is.) For example, in each of the following pairs the first formula is a theorem but the second is not:

- (1a) $(p \supset q) \vee (q \supset p)$
- (1b) $(p \rightarrow q) \vee (q \rightarrow p)$
- (2a) $(p \wedge q) \supset (p \supset q)$
- (2b) $(p \wedge q) \supset (p \rightarrow q)$

Moreover, sometimes we have an equivalence which is a theorem, but when $\supset$ is replaced by $\rightarrow$ the resulting formula is provable as an implication only. For example,

$$(3a) ((p \supset r) \vee (q \supset r)) \equiv ((p \wedge q) \supset r)$$

is a theorem, but while

$$(3b) ((p \rightarrow r) \vee (q \rightarrow r)) \supset ((p \wedge q) \rightarrow r)$$

is also a theorem, its converse is not. Here are some further theorems involving strict implication. They are numbered in sequence with the K theorems in Chapter 2.

$$\mathbf{K8} \quad (\sim p \rightarrow p) \equiv Lp$$

PROOF

- PC (1) $(\sim p \supset p) \equiv p$
- (1) $\times$ DR2 (2) $L(\sim p \supset p) \equiv Lp$
- (2)Def $\rightarrow$ (3) $(\sim p \rightarrow p) \equiv Lp$

Q.E.D.

Just as whenever we have $\vdash \alpha \supset \beta$ we also have $\vdash \alpha \rightarrow \beta$, so whenever we have $\vdash \alpha \equiv \beta$ we also have $\vdash \alpha = \beta$. I.e., K8 and all other equivalential theorems are also provable as strict equivalences.

$$\mathbf{K9} \quad (p \rightarrow \sim p) \equiv L\sim p$$

The proof is similar to that for K8.

$$\mathbf{K10} \quad ((q \rightarrow p) \wedge (\sim q \rightarrow p)) \equiv Lp$$

PROOF

- PC (1) $((q \supset p) \wedge (q \supset \sim p)) \equiv p$
- (1) $\times$ DR2 (2) $L((q \supset p) \wedge (\sim q \supset \sim p)) \supset Lp$
- (2)K3[$q \supset p/p, \sim q \supset p/q$] $\times$ Eq:

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$$(3) \text{Def } \rightarrow \quad (3) \quad (L(q \supset p) \wedge L(\sim q \supset p)) \equiv Lp \quad \text{Q.E.D.}$$

$$(4) \quad ((q \rightarrow p) \wedge (\sim q \rightarrow p)) \equiv Lp$$

$$\mathbf{K11} \quad ((p \rightarrow q) \wedge (p \rightarrow \sim q)) \equiv L\sim p$$

Proof as for K10.

K8-K11 express important facts about non-contingent propositions (i.e. propositions which are either necessary or impossible). K8 says that a necessary proposition is one which is strictly implied by its own negation. K9 says that an impossible proposition is one which strictly implies its own negation. K10 says that a necessary proposition is one which is strictly implied both by another proposition and by the negation of that other proposition. K11 says that an impossible proposition is one which strictly implies both another proposition and the negation of that other proposition.

$$\mathbf{K12} \quad Lp \supset (q \rightarrow p)$$

PROOF

$$\text{PC} \quad (1) \quad p \supset (q \supset p)$$

$$(1) \times \text{DR1} \quad (2) \quad Lp \supset L(q \supset p)$$

$$(2)\text{Def } \rightarrow \quad (3) \quad Lp \supset (q \rightarrow p) \quad \text{Q.E.D.}$$

$$\mathbf{K13} \quad L\sim p \supset (p \rightarrow q)$$

Proof as for K12.

K12 and K13 should be compared with (1) and (2) on p. 194. We will come back to them on pp. 202-204, since they have been the occasion of a large amount of controversy, but our immediate task is to return to Lewis's development of modal logic.

The 'Lewis' systems

In his early articles Lewis sometimes took strict disjunction as primitive, sometimes strict implication, sometimes logical impossibility; and in his book *A Survey of Symbolic Logic*,⁹ he set out an axiomatic system (the *Survey* system) in which he again took logical impossibility as the primitive modal operator (along with conjunction and negation as primitive truth-functional operators). In 1930 Oskar Becker¹⁰ proposed some additional axioms for the *Survey* system and showed that they enable all modalities to be reduced (see p. 52) to a small number of non-

equivalent ones. But the first comprehensive treatment of systems of strict implication (or indeed of systems of modal logic at all) appeared in 1932 in Lewis and Langford's book *Symbolic Logic*. Here possibility is taken as the primitive modal operator, and two axiomatic systems of strict implication (called S1 and S2 respectively) are developed in considerable detail. In an appendix several other systems are outlined as well: one of these is the system of the Survey (S3); two others, which contain certain of Becker's reduction postulates, are called S4 and S5.

Since Lewis assumed that what is necessary is true it is not to be expected that K would be one of his systems, but in fact T is not either. Nevertheless for purely first-degree wff, of the kind we have just been discussing, there is no difference between T and any of the Lewis systems. We shall set out these systems in the form in which they occur in *Symbolic Logic*, except that we shall use the notation and terminology employed in Chapter 1 of this book.¹¹

The system S1

*Primitive symbols*¹²

$p, q, r, \dots$	[Propositional variables]
$\sim, M$	[Monadic operators]
$\wedge$	[Dyadic operator]
$(,)$	[Brackets]

Formation rules

1. A propositional variable is a wff.
2. If α is a wff, so are $\sim\alpha$ and $M\alpha$.
3. If α and β are wff, so is $(\alpha \wedge \beta)$.

*Definitions*¹³

[Def $\vee$]	$(\alpha \vee \beta) =_{df} \sim(\sim\alpha \wedge \sim\beta)$
[Def $\rightarrow$]	$(\alpha \rightarrow \beta) =_{df} \sim M(\alpha \wedge \sim\beta)$
[Def $=$]	$(\alpha = \beta) =_{df} ((\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha))$
[Def L]	$L\alpha =_{df} \sim M\sim\alpha$

*Axioms*¹⁴

- AS1.1 $(p \wedge q) \rightarrow (q \wedge p)$
- AS1.2 $(p \wedge q) \rightarrow p$
- AS1.3 $p \rightarrow (p \wedge p)$
- AS1.4 $((p \wedge q) \wedge r) \rightarrow (p \wedge (q \wedge r))$
- AS1.5 $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

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AS1.6 $(p \wedge (p \rightarrow q)) \rightarrow q$

Transformation rules

1. *Uniform Substitution*, as in the systems in Part I.
2. *Substitution of strict equivalents*: If $\vdash \alpha$, and β differs from α only in having some wff, δ , at one or more places where α has a wff γ , then if $\vdash \gamma = \delta$, $\vdash \beta$.
3. *Adjunction*: $\vdash \alpha, \vdash \beta, \rightarrow \vdash \alpha \wedge \beta$.
4. *Modus Ponens (Detachment)*: $\vdash \alpha, \vdash \alpha \rightarrow \beta, \rightarrow \vdash \beta$.

There is one striking difference between the above basis for S1 and any of the bases discussed in earlier chapters, and that is that it is not constructed as an extension of PC.¹⁵ In fact none of the axioms of S1 is a wff of PC at all. Moreover, while the rule of Uniform Substitution belongs to PC, the S1 Modus Ponens rule is stated for strict implication, not for material implication as for PC. (We shall often call it the rule of Strict Detachment, and the corresponding PC rule, the rule of Material Detachment.) As a result proofs of theorems in S1 are apt to have a somewhat different ‘style’ from those in, say, T, since we are not free to help ourselves to any theorem of PC which seems likely to be useful. Nevertheless, S1 contains PC; i.e., every theorem of PC is a theorem of S1. It is easy to introduce the operators $\supset$ and $\equiv$ (as Lewis himself does¹⁶) by the definitions:

[Def $\supset$] $(\alpha \supset \beta) =_{df} \sim(\alpha \wedge \sim\beta)$

[Def $\equiv$] $(\alpha \equiv \beta) =_{df} ((\alpha \supset \beta) \wedge (\beta \supset \alpha))$

The axiom AS1.6 is interdeducible with $Lp \rightarrow p$. If it is omitted we have a system called S1⁰, which stands to S1 rather as K stands to T.¹⁷ In comparing S1 with T we first notice that the basis of S1 is certainly contained in T. Further, S1 contains the whole of the basis of T *except* for the rule of necessitation. In fact S1 has no theorems of the form $LL\alpha$ at all, and if so much as one is added the other rules enable the derivation of N, and increase S1 to T.¹⁸

Lemmon’s basis for S1

It is in fact possible to axiomatize S1 by making additions to non-modal PC as we did in earlier chapters. The following basis is due to E.J. Lemmon.¹⁹ Lemmon’s basis for S1 consists of the following axioms:

- (1) every PC-tautology;
- (2) $Lp \supset p$;
- (3) $(L(p \supset q) \wedge L(q \supset r)) \supset L(p \supset r)$.

The transformation rules are Uniform Substitution, Modus Ponens (for $\supset$), and two extra rules. The first is a restricted form of Necessitation:

N' If α is a PC-tautology or an axiom then $\vdash L\alpha$.

The second is a rule for the substitution of proved strict equivalents:

Eq' If α differs from β only in having a wff γ in some of the places where β has δ , and $\vdash \gamma = \delta$, then $\vdash \alpha = \beta$.

The system S2

S1 was not in fact Lewis's preferred system. The system he designated as the correct system is one called S2, obtained from S1 by adding

AS2.1 $M(p \wedge q) \rightarrow (Mp \wedge Mq)$

Lewis calls this the *Consistency Postulate*. Its sense is that only a possible (or *consistent*) proposition can be a term in a consistent conjunction. S2 can also be axiomatized in the style of Lemmon. We replace (3) with the wff **K** from p. 25

K $L(p \supset q) \supset (Lp \supset Lq)$.

We keep **N'** (now of course applied to **K** rather than (3)) and we replace **Eq'** with a rule called *Becker's Rule*.²⁰

BR $\vdash L(\alpha \supset \beta) \rightarrow \vdash L(L\alpha \supset L\beta)$

(Using $\rightarrow$ BR can be written as $\vdash \alpha \rightarrow \beta \rightarrow \vdash L\alpha \rightarrow L\beta$.)

The system S3

Although Becker's rule belongs to S2 the formula

(4) $(p \rightarrow q) \rightarrow (Lp \rightarrow Lq)$

which might be confused with it, is not a theorem of S2. Nevertheless it

could be added to S2, and if it is we obtain a system deductively equivalent to the system Lewis presented in his 1918 book. This system is called S3.²¹

Lewis also discussed S4 and S5. Although axiomatized differently, these systems are deductively equivalent to the S4 and S5 studied in Part I of this book.

Validity in S2 and S3

One reason why we shall have to make a substantial change in our earlier definitions of validity if we are to deal with S2 and S3 is that these definitions — i.e. those for T, S4, S5 and the like — all satisfy the rule of Necessitation. That is to say, if any wff, α , is valid in terms of any of these definitions, $L\alpha$ is also valid. But as we have observed, the rule of Necessitation does not hold, at least unrestrictedly, in S2 and S3.

Another, related, feature of S2 and S3 is that they are compatible with (though they do not contain) the axiom MMp .²² This means that they are compatible with (though they do not commit us to) the view that every proposition is ‘possibly possible’. And this suggests an idea which is in fact the key to S2- and S3-models, that there might be some ‘worlds’ in which every proposition without exception — even one of the form $p \wedge \sim p$ — is possible. Kripke²³ calls such worlds *non-normal* worlds, and we shall follow him in this terminology. Worlds of the kind that occur in the frames of normal modal systems are by contrast called *normal* worlds. The rules for evaluating non-modal formulae in non-normal worlds are the same as in normal worlds — thus even in a non-normal world we never have $p \wedge \sim p$ true for example — but for modal formulae in non-normal worlds $M\alpha$ is always true and $L\alpha$ is always false. In this respect non-normal worlds are the reverse of dead ends in normal modal logics.

In an S2 frame there must be at least one normal world, and there may (but need not) be one or more non-normal worlds.²⁴ Every normal world can see itself, and every non-normal world can be seen by at least one normal world. Otherwise the accessibility relations can be as we please. In an S3 frame there is the additional requirement that the accessibility relation be transitive. A formula will be said to be S2-(S3-)valid iff it is true in every *normal* world in every model based on an S2- (S3-) frame.²⁵

More exactly expressed, an S2 frame²⁶ is a triple $\langle W, R, N \rangle$ where W is a set of objects (worlds), N is a proper subset of W , i.e. $N \subseteq W$ but $N \neq W$, and R is a relation such that (a) R is reflexive over N , i.e. if $w \in N$ then wRw , and (b) for every $w' \in W$ there is some $w' \in N$ such

that wRw' . $\langle W, R, N, V \rangle$ is an S2-model iff $\langle W, R, N \rangle$ is an S2 frame and V is a value-assignment as on p. 38, except that [VL] should be changed to read that for any wff, α , and for any $w \in W$, $V(L\alpha, w) = 1$ if $w \in N$ and for every w' such that wRw' , $V(\alpha, w') = 1$. Otherwise $V(L\alpha, w) = 0$. (The effect of this is that if w is normal, $V(L\alpha, w)$ is computed as in a T-model; but if w is non-normal, $V(L\alpha, w) = 0$ in every case — and hence, incidentally, $V(M\alpha, w) = 1$ in every case.)

A wff α is *valid* on an S2 frame $\langle W, R, N \rangle$ iff in every S2-model $\langle W, R, N, V \rangle$ based on $\langle W, R, N \rangle$, $V(\alpha, w) = 1$ for every $w \in N$. A wff is S2-valid iff it is valid on every S2 frame. An S3 frame is defined in the same way as an S2 frame, except that we add the condition that R is transitive. A wff α is S3-valid iff it is valid on every S3 frame.

For those who prefer the approach via the parlour games of Chapter 1, the S2-game is the modal game with the following modifications. Some of the sheets of paper are, say, white, others pink. In every S2-setting at least one player must have a white sheet; but some of the players may have pink sheets instead. No player with a pink sheet may see any other player, but every such player must be seen by at least one player with a white sheet. The rules for responding to calls are, for players with white sheets, exactly as in the modal game. For players with pink sheets, rules 1–3 are as in the modal game, but rules 4 and 5 (those covering calls with L and M) are replaced by the following:

- 4'. If a call is of the form $L\alpha$, do not raise your hand.
- 5'. If a call is of the form $M\alpha$, raise your hand.

A call is an S2-successful call iff in every S2-setting it would lead every player with a white sheet to raise his or her hand. A formula is S2-valid iff it would form an S2-successful call. The S3-game will be the S2-game with the added rule that in every setting the seeing-relation must be transitive. S3-successful calls and S3-validity are then defined as above, with 'S3' replacing 'S2' throughout.

S2 and S3 may be shown to be sound and complete with respect to this semantics.²⁷ The soundness result also enables us to establish that they are distinct systems, and that neither contains S4. T contains S2 but neither contains nor is contained in S3. S1 is not susceptible of this kind of treatment and the only known semantics for it is unintuitive.²⁸

Entailment

An important modal notion is that of entailment. By this we understand

the converse of the relation of *following logically from* (when this is understood as a relation between *propositions*, not wff) i.e. to say that a proposition, p , entails a proposition, q , is simply an alternative way of saying that q follows logically from p , or that the inference from p to q is logically valid.²⁹ It is clear from the writings we have already referred to that Lewis wished to interpret $\rightarrow$ as 'entails'. Now there has been a good deal of philosophical controversy about the correct analysis of entailment and in particular K12 and K13 on p. 197 are sometimes known as the 'paradoxes of strict implication',³⁰ and are often considered problematic when $\rightarrow$ is interpreted as 'entails'. We shall look at them and some associated formulae in the following forms:

- (1) $(p \wedge \sim p) \rightarrow q$
- (2) $q \rightarrow (p \vee \sim p)$
- (3) $\sim Mp \supset (p \rightarrow q)$
- (4) $Lq \supset (p \rightarrow q)$ ³¹

When $\rightarrow$ is interpreted as 'entails', (1) means that from any proposition of the form $(p \wedge \sim p)$ any proposition whatever can be deduced, and (2) means that from any proposition whatever there can be deduced any proposition of the form $(p \vee \sim p)$. (3) and (4) are more general: (3) means that from any logically impossible proposition (whether of the form $(p \wedge \sim p)$ or not) any proposition whatever can be deduced, and (4) means that every necessary proposition (whether of the form $(p \vee \sim p)$ or not) can be deduced from any proposition whatever.

If these are not sound principles of deducibility, that would of course tell against the claim of the standard modal systems to be correct logics of entailment. But in order to decide whether they are sound principles of deducibility or not we have to look into what we take ourselves to be asserting when we assert that one proposition is deducible from another.

Now one plausible account is that to say that q is deducible from p is to say that it is logically impossible for p to be true but q false. Deducibility is after all the relation which obtains between the conclusion and the premiss(es) of a valid deductive inference, and what we require in a valid inference is the logical guarantee that we shall not have the premiss(es) true but the conclusion false. Now by this account the 'paradoxes' are sound principles of deducibility; and hence it is not their presence in but their absence from a system which would tell against its claim to be a correct logic of entailment. To take the case of (1): to say that $(p \wedge \sim p)$ entails q is on this account to say that it is logically

impossible for $(p \wedge \sim p)$ to be true but q false, i.e. it will amount to saying that $(p \wedge \sim p \wedge \sim q)$ is logically impossible; but since $(p \wedge \sim p)$ is itself impossible, so is $(p \wedge \sim p \wedge \sim q)$. Similar comments will apply to the other 'paradoxes'. Moreover, this account will guarantee that $\rightarrow$ can be interpreted as 'entails'; for in all the standard systems $\alpha \rightarrow \beta$ is defined as $\sim M(\alpha \wedge \sim \beta)$ (or, what comes to the same thing, as $L(\alpha \supset \beta)$), where M is interpreted as 'it is logically possible that'.

No one is likely to deny that the logical impossibility of $(p \supset q)$ is a necessary condition of q 's deducibility from p , but it has been suggested that it is not a sufficient condition on the ground that a further condition of q 's deducibility from p is that there should be some connection of 'content' or 'meaning' between p and q . But even those who are inclined to accept this further requirement for deducibility, however, have to face the following argument. On any account we shall have to regard q as deducible from p when it can be derived from p by some valid principle or principles of deductive inference. Now the following principles seem intuitively to be valid:³²

- A. Any conjunction entails each of its conjuncts.
- B. Any proposition, p , entails $(p \vee q)$, no matter what q may be.
- C. The premisses $(p \vee q)$ and $\sim p$ together entail the conclusion q (the principle of the disjunctive syllogism).
- D. Whenever p entails q and q entails r , then p entails r (the principle of the transitivity of entailment).

C.I. Lewis has shown that by using these principles we can always derive any arbitrary proposition, q , from any proposition of the form $(p \wedge \sim p)$, in the following way:

- | | |
|----------------------------|-----------------------|
| | (i) $p \wedge \sim p$ |
| From (i), by A: | (ii) p |
| From (ii), by B: | (iii) $p \vee q$ |
| From (i), by A: | (iv) $\sim p$ |
| From (iii) and (iv), by C: | (v) q ³³ |

By D we then have the result that $(p \wedge \sim p)$ entails q . This derivation shows that the price which has to be paid for denying that $(p \wedge \sim p)$ entails q is the abandonment of at least one of A–D.

The most fully developed formal response to these 'paradoxes' consists

of abandoning C, the principle of disjunctive syllogism. Logics which do this are called *relevance logics* and there is now an enormous body of literature on them.³⁴ These systems are well beyond the scope of the present book, and in fact relevance logics differ from all the logics we have so far considered in that they require a non-standard interpretation of the PC symbols, in particular of negation.

Exercises — 11

11.1 Prove that adding $LL(p \supset p)$ to S1 gives T.

11.2 Prove that Lemmon's basis for S1(S2) on pp. 199–200 is equivalent to Lewis's.

11.3 Where S7 is $S3 + MPp$, prove that $\vdash_{S3} \alpha$ iff $\vdash_{S4} \alpha$ and $\vdash_{S7} \alpha$.

11.4 Let E2 be $\{\alpha: L\alpha \in S2\}$.

(a) Prove that N is not a rule of E2.

(b) Prove that E2 is sound with respect to the class of S2 frames but with the definition of validity changed so that a wff is E2-valid iff it is true in *all* worlds, not just normal worlds, in every frame.

(c) Give completeness proofs for E2 and S2 by defining a canonical model in which W is the set of all maximal E2-consistent sets of wff and $w \in N$ iff $L(p \supset p) \in w$.

11.5 Show that E2 can be axiomatized by PC, T, K, and the rules US, MP and R*: $\vdash \alpha \supset \beta \rightarrow \vdash L\alpha \supset L\beta$.

11.6 Where E3 is E2 with $L(p \supset q) \supset L(Lp \supset Lq)$ show that E3 is characterized by S3 frames when validity is truth in all worlds in every S3 frame.

11.7 S3.5 is $S3 + Mp \supset LMp$ (see note 27). Where an S3.5 frame is an S3 frame in which R is symmetrical over N (i.e. if $w, w' \in N$ and wRw' then $w'Rw$) show that S3.5 frames characterize S3.5.

11.8 S0.5 is just like T except that N is replaced by the rule:

N' If α is a PC-valid wff then $\vdash \alpha$.

A model for S0.5 consists of a set of worlds, of which one, w^* , is a 'distinguished' world. For w^* , $V(L\alpha, w^*) = 1$ iff $V(\alpha, w) = 1$ for every $w \in W$. For every other world $L\alpha$ has an arbitrary value. A wff α is

S0.5-valid iff $V(\alpha, w^*) = 1$ in every S0.5 model.

- (a) Prove that S0.5 is sound with respect to this definition of validity.
- (b) Prove that N is not a rule of S0.5.
- (c) Prove that Eq is not a rule of S0.5.

11.9 Construct a canonical model for S0.5 in which w^* is a set of maximal S0.5-consistent sets of wff and every other world is a maximal PC-consistent set of wff (i.e. w is maximal and there is no set $\{\alpha_1, \dots, \alpha_n\}$ such that each of $\alpha_1, \dots, \alpha_n$ is in w and $\sim(\alpha_1 \wedge \dots \wedge \alpha_n)$ is a substitution-instance of a PC-tautology). Use this to prove the completeness of S0.5.

Notes

¹ Aristotle, 350 BC, 29b29–40b16. An attempt to formalize Aristotle's modal logic will be found in McCall 1963. For a general history of ancient and mediaeval modal logic see Kneale and Kneale 1962, pp. 81–96, 117–138, 212, 232, 236, 243, or Bocheński 1961, pp. 81–88, 101–103, 114–115, 224–230.

² MacColl 1880, pp. 50–55.

³ MacColl 1903, 1906a, 1906b, (see especially 1903, pp. 356–7.

⁴ He does give (1906a, p. 8) a list of 'self-evident formulae' and it would be interesting to know which of the more recent modal systems is the weakest in which all these are true.

⁵ Whitehead and Russell 1910.

⁶ Lewis 1912, 1913, 1914a, 1914b, 1918. Lewis and Langford 1932.

⁷ As far as we have been able to discover, the term 'strict implication' first occurs in Lewis 1912, p. 526 n. 1. The symbol $\supset$ appears in Lewis 1918.

⁸ = as defined here and $=_{df}$ should not be confused. The former is an operator which occurs in wff of a modal system; the latter is a metalogical symbol which never occurs in wff but is used only in discoursing about a system.

⁹ Lewis 1918, ch. 5 (emended in Lewis 1920).

¹⁰ Becker 1930.

¹¹ These names ('S1' etc.), by which the systems have since become generally known, are given on pp. 500–501 of Appendix II (written by Lewis) in Lewis and Langford 1932. They do not occur in Chapter 6, where S1 and S2 are developed (unless we count a brief reference to 'System 1' and 'System 2' on pp. 177–178). Lewis's S4 and S5 are deductively equivalent to the S4 and S5 of Part I of this book, though they have different bases. S4 and S5 appear in Lewis and Langford 1932 only in the appendix on p. 500f., and are rejected by Lewis as acceptable systems of strict implication. S3 is the system of Lewis 1918, and S1 and S2 are the systems developed in Chapter 6 of Lewis and Langford 1932. For a more detailed survey of the axioms, theorems and rules of the various Lewis systems see Feys 1965, Chapters 12 and 13 of Hughes and Cresswell 1968, and Zeman

1973.

¹² Lewis uses $\sim$ for negation, juxtaposition for conjunction, and $\diamond$ for M .

¹³ We write these definitions in the style adopted in Part I. Lewis writes them as strict equivalences, using propositional variables. Lewis does not have a single symbol for necessity, but writes $\sim \diamond \sim$ throughout. (His $\diamond =$ our M). However, the abbreviation provided by this definition is an obvious convenience.

¹⁴ Our numbering of these axioms is not the same as that of Lewis and Langford. Moreover we omit the axiom $p \supset \sim \sim p$ since this was shown to be non-independent in McKinsey 1934. Instead of AS1.6 we may have $\sim Mp \supset \sim p$, or $p \supset Mp$.

¹⁵ The first axiomatization of modal logic starting from a PC basis and adding extra axioms and rules to it (as in Part I) appears to be that in Gödel 1933.

¹⁶ Lewis and Langford 1932, p. 13ff.

¹⁷ Feys 1965, p. 43.

¹⁸ Yonemitsu 1955.

¹⁹ Lemmon 1957. Lemmon also considers a weaker system, which he calls S0.5, in which (3) is replaced by **K**, and **N'** by the rule **N''** that if α is a PC-tautology then $\vdash L\alpha$. Interestingly S0.5 does not satisfy the rule Eq, even for proved strict equivalents. (See Hughes and Cresswell 1968, pp. 286–288.) By omitting $Lp \supset p$ we obtain S0.5⁰. In Lemmon 1959, p. 31, there is the suggestion that in S0.5 the necessity operator might mean ‘it is tautologous by truth tables that ...’.

²⁰ Becker 1930. The name ‘Becker’s Rule’ was given in Churchman 1938.

²¹ S3 was subsequently discovered to be stronger than S2, but Lewis in 1932 (p. 496) had no proof of this and declared that if S2 should turn out to contain S3 he would fall back on S1, which he knew to be weaker than S2. Parry 1939 proves that S3 has only 42 distinct affirmative modalities. The systems which result from S3 by adding all possible modality reduction laws are classified in Pledger 1972 and given a possible-worlds semantics in Goldblatt 1973.

²² This wff is called C13 on p. 497 of Lewis and Langford 1932.

²³ Kripke 1965b, p. 208 uses a slightly different axiomatization based on an infinite (though effectively specifiable) set of axioms with material detachment as the only primitive rule of inference. His axiomatic basis may be easily shown equivalent to Lemmon’s and for our purposes there is nothing to choose between them though, unlike the bases we are using, Kripke’s basis allows the addition of $LL(p \supset p)$ to S2 without obtaining the unrestricted rule of Necessitation and permits an infinity of systems to be generated by the axioms $L^n(p \supset p)$ (for each n). For some suggestions for interpreting S2 see Cresswell 1967b. A canonical model completeness theorem for S2 appears in Cresswell 1982.

²⁴ If we insist that there must be at least one non-normal world then MMp becomes valid. $S2 + MMp$ has been called S6 (Alban 1943) and $S3 + MMp$ S7. (Halldén 1949a, Hughes and Cresswell 1968, pp. 281–284.) S3 is the intersection of S4 and S7. $S3 + LMMp$ is called S8. In S8 frames every normal world can see a non-normal world.

²⁵ If we define validity as truth in *all* worlds we get a semantics for the 'E-systems' of Lemmon 1957 (see Hughes and Cresswell 1968, pp. 302f). In these systems N is replaced by the rule $R^* \vdash \alpha \supset \beta \rightarrow \vdash L\alpha \supset L\beta$. Unlike normal systems, which contain N, these systems have no theorems of the form $L\alpha$, and their canonical models contain maximal consistent sets with no wff of that form. Segerberg 1971, Chapter 4, calls such systems 'regular' and calls systems like S2 and S3 'quasi-regular'. That Chapter shows how to apply techniques from normal modal logic to regular and quasi-regular logics. One can also study logics in which all worlds are normal but in which validity is defined as truth in a designated subset of worlds. Chapter 3 of Segerberg 1971 calls these 'quasi-normal' systems. They all contain (all the theorems of) K, and the rules US and MP, but not the rule of necessitation. An example of a quasi-normal system is studied in Langholm 1987. It is $K + p \supset L^nMp$ and is intended to formalize a system of logic advocated in Smith 1936.

²⁶ Kripke 1965b does not take N as primitive but defines it via R. He calls R 'quasi-reflexive' provided that for any $w, w' \in W$, if wRw' then wRw (i.e. a world which can see anything can see itself) and then defines a world as normal iff wRw . We have used N to make the semantics easier to follow. (Also using N generalizes more easily to systems where R is not reflexive over normal worlds.)

²⁷ Where a normal system S is characterized by a class of frames there is of course the non-normal system characterized by the class of all frames obtained from frames for S by the addition of non-normal worlds with the condition that every non-normal world can be seen by a normal world. All such systems will be extensions of the system Feys 1950, 1965, p.68, calls $S2^0$, i.e., in Lemmon's axiomatization, S2 without $Lp \supset p$. This system is characterized by frames in which no restrictions are placed on R, except that every non-normal world can be seen by a normal world. $S2^0$ corresponds to K as S2 corresponds to T and S3 to S4. Corresponding to S5 is a system called S3.5, which is obtained by adding E to S3 (Åqvist 1964). Note that the strict form of E, $L(Mp \supset LMp)$ strengthens S3 to S5. A completeness theorem for S3.5 is found in Cresswell 1967a. For S3.5 we may also prove a conjunctive normal form theorem (Cresswell 1969a). The system corresponding to B is $S2 + B$. ($S3 + B$ is S3.5.) $S3.5 + MMP$ has been called S9. (See Hughes and Cresswell 1968, pp.172 and 285f.)

²⁸ Cresswell 1995a.

²⁹ This use of 'entails' has for some time been standard in philosophy. It derives from Moore 1919 (reprinted in his *Philosophical Studies*; see esp. p. 291). It is important at this point to stress that we are here thinking of a relation between propositions rather than wff. For there is a quite different, though equally legitimate, use of the term 'logically follows from', whereby a wff β 'logically follows from' a wff α in a logical system S iff $\vdash_S \alpha \supset \beta$. The distinction between these two senses of 'logically follows from' parallels the distinction between validity and necessary truth. We shall have more to say on this on p. 225.

STRICT IMPLICATION

³⁰ Tendentiously; for those on the other side in the controversy regard the formulae as expressing perfectly sound principles of deducibility, and on anyone's account they express sound and quite unparadoxical truths about strict implication. The 'paradoxes' seem to have been first stated (and incidentally accepted as unparadoxical) in modern logic by MacColl 1906b, p. 613. For some information about mediaeval anticipations of them see Kneale and Kneale 1962, pp. 281ff.

³¹ In S2 and stronger systems we can also prove $\sim Mp \supset (p \supset q)$ and $Lq \supset (p \supset q)$. Two early attempts to formalize a relation which does not lead to the 'paradoxes' (Emch 1936 and Vredenduin 1939) avoided them in the S2 forms but contained our (1) and (2) as theorems.

³² To say that q may be derived from p by some valid principle(s) of inference is (as noted in Lewis and Langford 1932, pp. 252–255) not the same as saying that it may be derived by the principles of a given system, or by principles we have already established up to a given point in the development of a system. Rather it is to say that the principles which enable us to pass from p to q are sound ones, whether they occur in any particular system or not. See Pollock 1966, pp. 184–185, for a discussion of this confusion in writers later than Lewis.

³³ Cf. Lewis and Langford 1932, pp. 250–251, where there is also found an analogous derivation, relevant in a similar way to our (2), of $\sim q \vee q$ from p . For a mediaeval anticipation of Lewis's derivation of q from $p \wedge \sim p$ see Kneale and Kneale 1962 and Kneale 1956, pp. 239–240. It is also possible to derive the result that $(p \wedge \sim p)$ entails $\sim q$ (a simple and equally general variant of the 'paradox' in question) by starting from the principle that $(p \wedge \sim q)$ entails p and applying to it the principle of antilogism, viz. that if $(p \wedge q)$ entails r then $(p \wedge \sim r)$ entails $\sim q$ (see Lewis 1914a, p. 246n, and Moh Shaw-Kwei 1950, p. 70).

³⁴ A survey of relevance logic is found in Dunn 1986.